# Linear Models Lecture 17: Pooled and Panel Data

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# From Cross-Sections to Repeated Observations

- A single cross-section identifies effects only by comparing units. Adding time lets us compare a unit to itself.
- Time can enter the model in several ways:
  - As an indicator (before/after a policy) or as a continuous trend.
  - Through interactions, allowing slopes to vary by period (the effect of a free fedora on employment may differ today from 1940).
- The simplest design exploits two groups at two times: **difference-in-differences**. We treat it as the on-ramp to panel data.

# Differences in Differences

Voter ID law in Texas; Oklahoma is the control. Turnout $Y$ observed before and after.

**Parameters:** $M$ = Texas baseline (e.g. borders Mexico), $O$ = Oklahoma baseline; $T$ = common period effect (e.g. Trump on the ballot), $D$ = effect of the voter ID law.

**Cross-section (after)**

| | |
|---|---|
| Texas | $M + T + D$ |
| Oklahoma | $O + T$ |
| Diff | $(M - O) + D$ |

**Time-series (Texas)**

| | |
|---|---|
| Before | $M$ |
| After | $M + T + D$ |
| Diff | $T + D$ |

Neither comparison alone recovers $D$.

# Difference-in-Differences: Recovering $D$

Difference each state across time, then difference the within-state differences.

## Difference-in-Differences

| | Before | After | After − Before |
|---|---|---|---|
| Texas | $M$ | $M+T+D$ | $T+D$ |
| Oklahoma | $O$ | $O+T$ | $T$ |
| | Texas − Oklahoma: | | $\boldsymbol{D}$ |

Within-state differences cancel $M$ and $O$; differencing them cancels $T$.

# Potential Outcomes

To define the assumption that lets DD work, restate the setup formally. $D$ stays the effect size; use $G$ for the group indicator.

- $G \in \{0, 1\}$: group indicator ($1 =$ treated group, e.g. Texas; $0 =$ control, e.g. Oklahoma).
- $t \in \{0, 1\}$: period ($0 =$ before, $1 =$ after).
- $Y_t^g$: potential outcome in period $t$ if group status were $g$.
- Observed outcome: $Y_t = G\, Y_t^1 + (1 - G)\, Y_t^0$.
- Two are observed, two are counterfactual:
    - $Y_t^1 \mid G = 0$: Oklahoma's outcome had it been treated.
    - $Y_t^0 \mid G = 1$: Texas's outcome had it not been treated.
- Unit-level treatment effect: $\delta_i = Y_t^1 - Y_t^0$. Across the population this averages to $D$.

# Parallel Trends

DD recovers $D$ only if Texas and Oklahoma would have moved together absent the law.

**Parallel trends assumption**

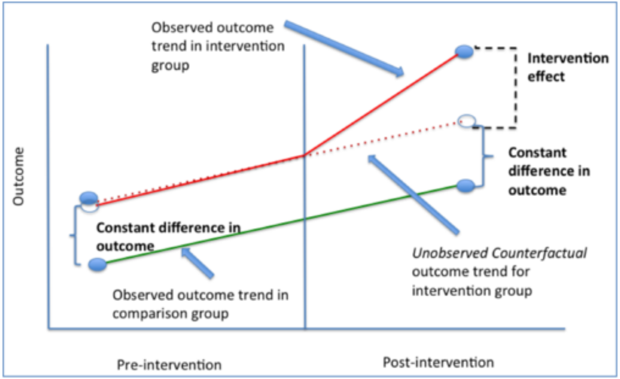
$$E[Y^0_{t=1} \mid G = 1] - E[Y^0_{t=0} \mid G = 1] \; = \; E[Y^0_{t=1} \mid G = 0] - E[Y^0_{t=0} \mid G = 0]$$

In the running example: $T$ enters both states the same way, and the baseline gap $M - O$ does not drift over time.

**Violations:**

- Different state-level trends (Texas suburbanizing faster than Oklahoma).
- Treatment-induced selection or composition change (voters moving out of Texas after the law).

# Parallel Trends, Visually




The dotted line is the counterfactual under parallel trends: it follows the comparison group's slope. The gap on the right is the treatment effect.

# Obama Support: The Identification Problem

- Suppose we want to study the effect of racial co-identification (non-identification) on feeling thermometers of Obama in 2008.

$$ObamaRating_i = b_0 + d_O\,Black_i + \mathbf{b}_{\mathbf{x}}'\mathbf{x}_i + e_i$$

- However, we are worried about the two kinds of omitted factors:
  - Time trends (T): Obama ran after a financial crisis and war that might make him popular generally.
  - Group Differences (M-O): Black and white people might differ in their preferences over any president that is a relatively liberal northern antiwar Democratic senator.
- Suppose Obama's appeal (repulsion) among Black (white) people depends on the economic conditions in 2008, his being a liberal Senator, and not only his race?

# Obama Support: DiD with a Comparison Candidate

- We can use a repeated cross section to difference out similar attitudes toward another Democratic candidate.

$$Kerry_i = b_0 + d_K\, Black_i + \boldsymbol{b}_{\boldsymbol{x}}' \boldsymbol{x}_i + e_i$$

- If we assume that
  - All time varying factors other than the candidate's race affected Black and white people's response to Democratic nominees equally.
  - All differences in evaluations of D. nominees between Black and white people did not change between the Kerry and Obama years.
- Then $\delta$ is an estimate of the Average Treatment Effect of the candidate's race on attitudes.

$$\delta = d_O - d_K = (\bar{r}_{O,B} - \bar{r}_{O,W}) - (\bar{r}_{K,B} - \bar{r}_{K,W})$$

# Estimating Differences in Differences

- Stack the Kerry and Obama waves. Define $yr08_i = 1$ for 2008 survey takers and $B_i = \mathbb{I}(\text{Race="Black"})$.
- **Unrestricted** (year-specific race effect):

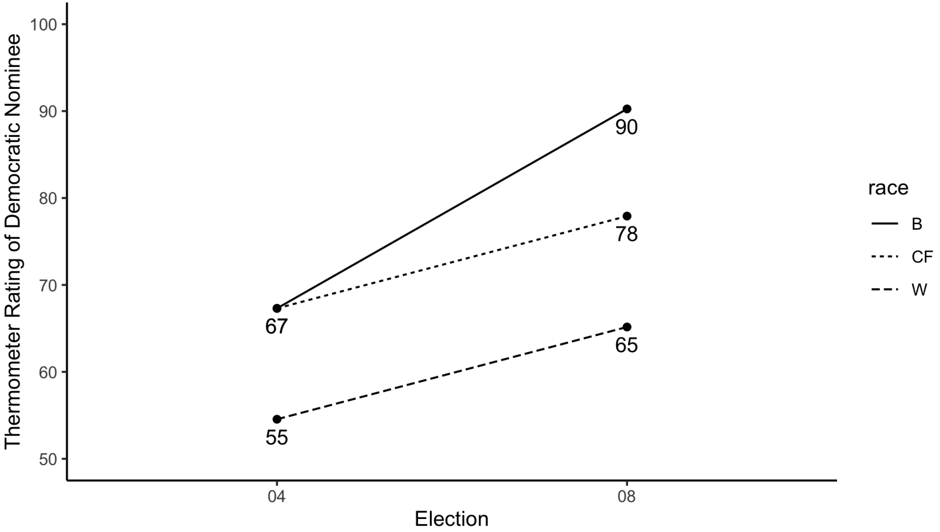
$$DRating_i = b_{0,O}\, yr08_i + b_{0,K}\, (1 - yr08_i) + d_O\, yr08_i\, B_i + d_K\, (1 - yr08_i)\, B_i + \boldsymbol{b}_x' \boldsymbol{x}_i + e_i$$

- **Restricted** ($d_O = d_K \equiv d$):

$$DRating_i = b_{0,O}\, yr08_i + b_{0,K}\, (1 - yr08_i) + d\, B_i + \boldsymbol{b}_x' \boldsymbol{x}_i + e_i$$

- The F test of $H_0\colon\ d_O = d_K$ comparing the two models is the DiD test: the racial co-identification effect is $d_O - d_K$.

| | demft | | |
|---|---|---|---|
| I(1 - y08) | 54.5*** | 53.0*** | |
| | (1.4) | (1.4) | |
| y08 | 65.2*** | 65.9*** | 10.6*** |
| | (1.3) | (1.3) | (1.0) |
| black | | 21.7*** | 12.8*** |
| | | (1.0) | (1.9) |
| I(1 - y08):black | 12.8*** | | |
| | (1.9) | | |
| y08:black | 25.1*** | | 12.3*** |
| | (1.2) | | (2.3) |
| age | 0.02 | 0.02 | 0.02 |
| | (0.02) | (0.02) | (0.02) |
| Constant | | | 54.5*** |
| | | | (1.4) |


Thermometer Rating of Democratic Nominee
100
90
80
70
60
50
90
78
67
65
55
04
08
Election
race
B
CF
W

# From Two Cells to Many: Panel Data

- DiD compared two groups at two times. Differencing cancelled time-invariant group features and group-invariant time trends.
- Panel data observes the *same* units repeatedly, so we can perform that cancellation for every unit, every period.
- Two costs of relaxing the two-cell setup:
  - Composition: groups must remain comparable across periods (no entry or exit driven by treatment).
  - Dependence: repeated observations on the same unit are correlated, breaking iid.
- The remaining sections build pooled, between, and within estimators on this richer structure.

# Three Examples to Keep in Mind

- Each pairs a time-varying outcome $Y_t$, a time-varying regressor $Z_t$, and an unobserved unit-level feature $A$ that we cannot measure directly.
- The same machinery (within transformation, GMM moment conditions) applies to all three, but the assumptions differ.
- As we proceed, ask of each example: which estimator handles it, and which would miss?

# Panel Example 1: Voting behavior

- Consider a population of families. Choose one at random.
- For each family member $t$, we observe voting decisions $Y_t$ and exposure to a persuasive messages $Z_t$.
- Each family has a household wealth $W$ and a set of stories passed down from their grandparents $A$.
- We observe $Y_t$, $Z_t$ and $W$, but not $A$.
- We want to estimate the effect of $Z_t$ on $Y_t$, holding constant $W$ and $A$.

# Panel Example 2: Airline cost structure

- Consider a population of airlines. Choose one at random.
- In year $t$, the airline incurs cost $Y_t$ to fly $Z_t$ passengers.
- Each airline has a fixed efficiency $A$ (technology, route network, labor contracts) that does not vary over time.
- We observe $N$ randomly sampled airlines but do not measure $A$.
- Costs scale with passengers, but different airlines spend different amounts to fly the same load. We want the within-airline cost-output elasticity, separating scale economies from efficiency $A$.

# Panel Example 3: Candidate Electoral Performance

- Consider a population of candidates. Choose one at random.
- In election cycle $t$, the candidate receives vote share $Y_t$.
- Each candidate has an underlying charisma $A$ (intrinsic appeal, base of supporters) that does not vary over time.
- We observe $N$ randomly sampled candidates across cycles, but do not measure $A$.
- Candidates with strong cycles often have strong follow-up cycles, partly because last cycle's success builds name recognition, donor networks, and incumbency benefits, and partly because they have higher charisma. We want the persistence of vote share, $Y_{t-1} \rightarrow Y_t$, separating that carryover from $A$.

# What These Examples Motivate

- Voting in families: $A$ (grandparents' stories) likely correlates with $Z$ (messages); pooled OLS is biased and **fixed effects** restore comparability.
- Airline costs: $A$ (proprietary technology) does not vary in time, so the **within** estimator removes it without measuring it.
- Candidate performance: $Y_{t-1}$ appears on the right-hand side; the within transformation correlates with the error and **dynamic panel GMM** (next lecture) instruments with lagged levels.

# Panel Data: Terminology

- The voters in a family, the airlines and candidates over time are all examples of *panel data*.
- Our observations are double indexed, usually divided into $N$ units and $T$ time periods/groups.
- If there are many $T$ but few $N$, we call the data long and narrow, (time series cross section).
- If every unit is observed in every period/ group, the data are called balanced.

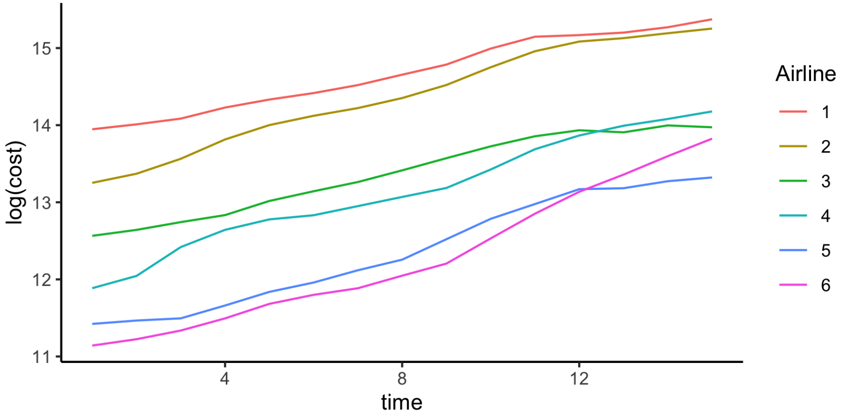
# Greene's Airline Panel

- Greene analyzes the cost structure in the airline industry from 1970-84.
- Research question: do airlines have positive economies of scale?
- library(AER); data("USAirlines"); analyzes determinants of total costs for 6 airlines.
- The cost function follows a Cobb-Douglas specification. Taking logs yields a linear regression:

$$\ln(\text{cost}) = \beta_0 + \beta_1 \ln(\text{output}) + \beta_2 \ln(\text{fuel price}) + \beta_3 \text{load} + u$$

- *cost*: total costs of labor materials etc in USD 1000.
- *output*: revenue passenger mile index
- input *prices*: fuel prices
- *load*: share of seats sold on flights flown.

# Airline Panel: Six Carriers, 15 Years




Log total cost, 1970–1984. Levels differ across carriers; all rise together over time.

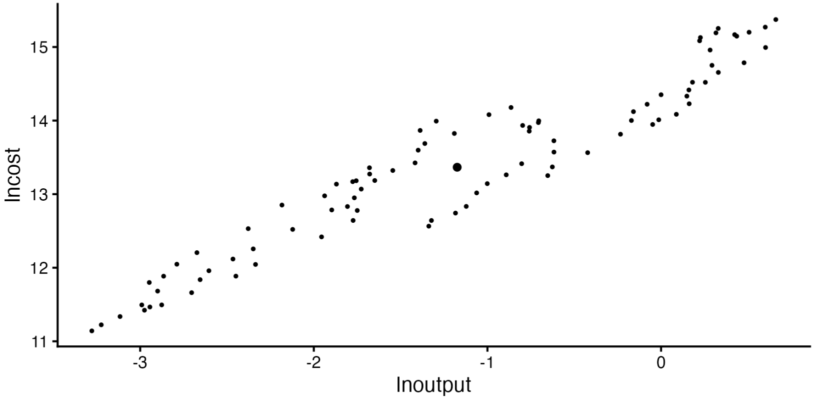
# Pooled Regression of log(costs)

| | Greene 2000 | Greene 2003 |
|---|---|---|
| (Intercept) | $9.52^{***}$ | $9.42^{***}$ |
| | (0.23) | (0.23) |
| lnoutput | $0.88^{***}$ | $0.94^{***}$ |
| | (0.01) | (0.03) |
| lnpfuel | $0.45^{***}$ | $0.46^{***}$ |
| | (0.02) | (0.02) |
| load | $-1.63^{***}$ | $-1.54^{***}$ |
| | (0.35) | (0.34) |
| $\text{lnoutput}^2$ | | $0.02^{*}$ |
| | | (0.01) |
| $R^2$ | 0.99 | 0.99 |
| Num. obs. | 90 | 90 |

# Where Does the Variation Come From?

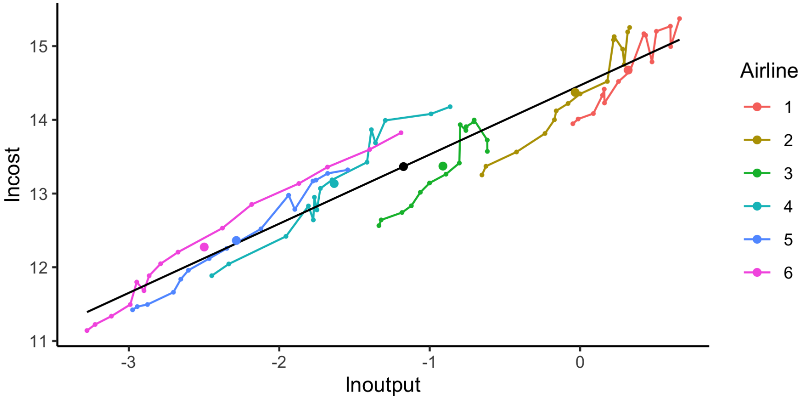
- Before reading off pooled coefficients, look at the data the way pooled OLS does: every airline-year is one point.
- Then strip off the airline-specific mean and ask what is left within an airline.
- The next plots compare those views for output, fuel price, and load.

# Cost and Output: Raw Scatter



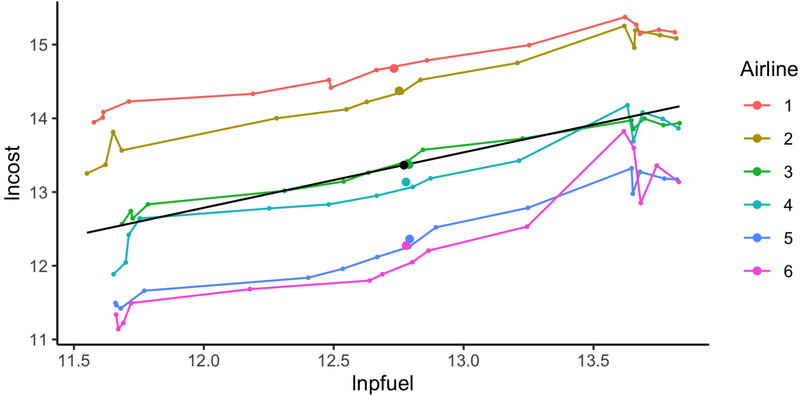

90 airline-years pooled. The black dot is the grand mean ($\overline{\overline{\text{lnoutput}}}$, $\overline{\overline{\text{lncost}}}$), the only summary a single OLS fit will use.

# Cost and Output: Pooled, Between, Within



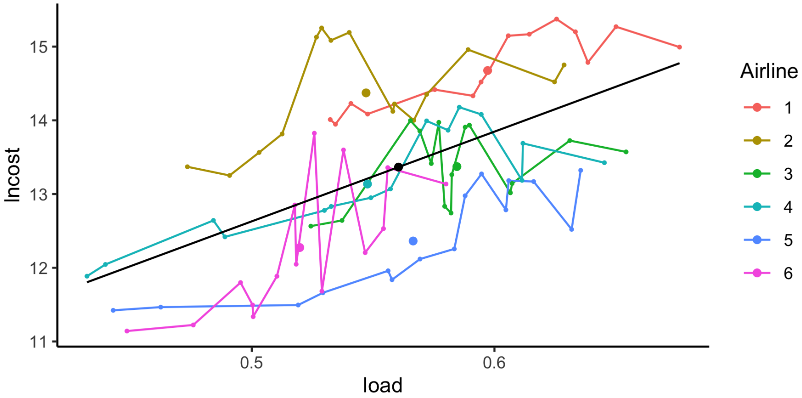

Coloured lines: each airline through time (within variation). Large coloured dots: airline means (between variation). Black line: pooled OLS fit.

# Cost and Fuel Price



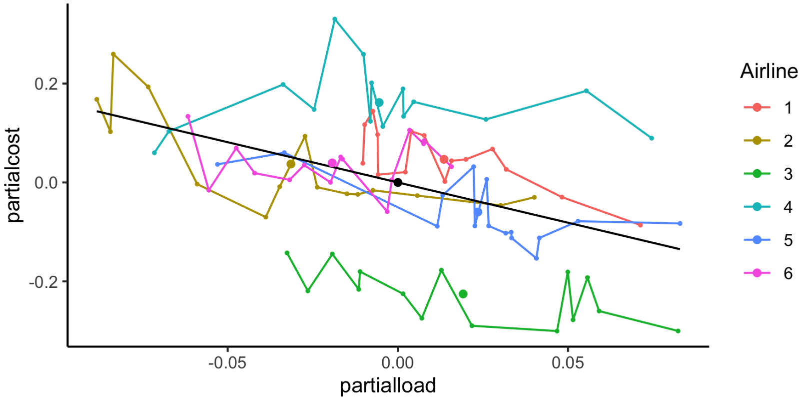

Within and between move the same way (prices rose over the 1970s for everyone), so the pooled slope is close to the within slope.

# Cost and Capacity Utilization (load)




Bivariate pooled slope is mildly *positive*: airlines that run fuller planes also have higher total costs, because larger carriers do both.

# Cost and Load, partialling out output and fuel price




Residual scatter from regressing both variables on (lnoutput, lnpfuel). Slope flips to $-1.63$: holding scale and input prices fixed, fuller planes are cheaper.

# What the Plots Show

- For output and fuel price, within and between push the same way, so pooled OLS looks fine.
- For load, between-airline variation reverses the sign relative to within. The bivariate pooled scatter is misleading.
- Partialling out the other regressors recovers the within-airline relationship. FE will do this for all variables at once.

# Assumptions of Pooled Regression: Exogeneity

- Before turning to fixed effects, ask what pooled OLS needs to be unbiased once the same unit appears $T$ times.
- $\boldsymbol{y} = \boldsymbol{X}\beta + \boldsymbol{e}$
- Strict Exogeneity: $E(e_{it}|\boldsymbol{X}) = \boldsymbol{0}$ for all time periods of $\boldsymbol{X}$.
- That is, $\boldsymbol{X}$ must not have any long-run consequences for $\boldsymbol{Y}$.
- Moreover, $\boldsymbol{X}$ must not itself be a function of past $\boldsymbol{Y}$.

# Autocorrelation versus Autoregression

$$Y_t = \alpha Y_{t-1} + \beta X_t + \nu_t$$
$$\nu_t = \rho \nu_{t-1} + e_t$$

- $\rho \neq 0$, $\alpha = 0$ has a "common factor" problem (autocorrelation):
  - Outcomes at $t-1$ are associated with outcomes at $t$ for many unobserved reasons.
  - OLS slope estimates are unbiased but inefficient and produces wrong standard errors.
  - Addressed with GLS/ serial correlation robust standard errors.
- $\alpha \neq 0$ is associated with dynamics:
  - Outcomes at $t-1$ affect outcomes now.
  - OLS is biased, inefficient and produces wrong standard errors.
  - Addressed with lagged variables and other structural time series approaches.

# Improvements on Inference

- In panel data we likely violate of iid errors, as observations from the same group/firm are likely associated
$$E[e_{ig}e_{jg}] = \rho\sigma_e^2 > 0$$
- Here $\rho$ is the intra-group/firm correlation.

# Standard error problems.

- If the number of groups is large, we can cluster standard errors using a formula similar to White standard errors.
- If the number of groups is small, we can just inflate our standard errors by $\sqrt{1+(n_g-1)\rho}$ (the Moulton factor; $n_g$ is the (average) cluster size).
- There is a bootstrap solution that has good performance

# White vs Cluster Robust

| | (white) | (cluster by airline) |
|---|---|---|
| (Intercept) | 9.52*** | 9.52*** |
| | (0.22) | (0.38) |
| lnoutput | 0.88*** | 0.88*** |
| | (0.01) | (0.02) |
| lnpfuel | 0.45*** | 0.45*** |
| | (0.02) | (0.03) |
| load | −1.63*** | −1.63* |
| | (0.32) | (0.44) |
| $R^2$ | 0.99 | 0.99 |
| N Clusters | | 6 |

# From Fixing Inference to Using the Panel Structure

- Clustered SEs adjust the variance of $\hat{\boldsymbol{b}}_P$ but leave the coefficient alone.
- The deeper problem is bias when $\alpha_i$ is correlated with $\boldsymbol{X}_{it}$: pooled OLS averages it in.
- Each unit is observed $T$ times. Differencing each variable from its own unit mean cancels $\alpha_i$ without ever observing it.
- That move defines the within (fixed-effects) estimator. We derive it next, then show that pooled OLS is itself a weighted average of within and between.

# Fixed Effects: Two Equivalent Approaches

Panel model $y_{it} = \boldsymbol{\beta}' \boldsymbol{x}_{it} + \alpha_i + e_{it}$, with $E[e_{it} \mid \alpha_i, \boldsymbol{x}_{i\cdot}] = 0$ but $E[\alpha_i \mid \boldsymbol{x}_{it}] \neq 0$ (so pooled OLS is biased).

### Dummy variables

Add one dummy per unit, run OLS:

$$\boldsymbol{y} = \boldsymbol{X}\beta + \boldsymbol{D}\boldsymbol{\alpha} + \boldsymbol{e}.$$

The $N$ fitted dummies absorb the unit intercepts.

### Demean within units, then regress

Subtract each variable's own unit mean and run OLS:

$$y_{it} - \bar{y}_i = \boldsymbol{\beta}'(\boldsymbol{x}_{it} - \bar{\boldsymbol{x}}_i) + \tilde{e}_{it}.$$

The $\alpha_i$ cancels because $\alpha_i - \bar{\alpha}_i = 0$.

**Both produce the same $\hat{\beta}$.** Why: Frisch-Waugh-Lovell turns the dummy regression into a regression on within-demeaned variables. The next three slides prove it.

# Why the Two Match: Frisch-Waugh-Lovell

By FWL, the slope on $\boldsymbol{X}$ in the dummy regression equals OLS run on $\boldsymbol{X}$ and $\boldsymbol{y}$ after partialling $\boldsymbol{D}$ out of both:

$$\hat{\beta} = (\boldsymbol{X}'\boldsymbol{M}_D\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{M}_D\boldsymbol{y}, \qquad \boldsymbol{M}_D = \boldsymbol{I} - \boldsymbol{D}(\boldsymbol{D}'\boldsymbol{D})^{-1}\boldsymbol{D}'.$$

$\boldsymbol{M}_D$ is the residual maker for the dummy regression: $\boldsymbol{M}_D\boldsymbol{v}$ is whatever is left of $\boldsymbol{v}$ after a regression on the unit dummies alone.

**Claim.** For any variable $\boldsymbol{v}$, $\boldsymbol{M}_D\boldsymbol{v}$ is $\boldsymbol{v}$ minus its unit mean: $(\boldsymbol{M}_D\boldsymbol{v})_{it} = v_{it} - \bar{v}_i$. Granted the claim, the dummy estimator runs OLS on the within-demeaned data. The next two slides verify the claim for $N = 2$.

# Computing $M_D$ for $N = 2$

- $\boldsymbol{D} = \begin{pmatrix} \boldsymbol{i} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{i} \end{pmatrix}$ where $\boldsymbol{i}$ is a $(T \times 1)$ vector of 1s.

$$\begin{aligned}
\boldsymbol{M}_D &= \boldsymbol{I} - \boldsymbol{D}(\boldsymbol{D}'\boldsymbol{D})^{-1}\boldsymbol{D}' \\
&= \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{pmatrix} - \begin{pmatrix} \boldsymbol{i} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{i} \end{pmatrix} \left[ \begin{pmatrix} \boldsymbol{i}' & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{i}' \end{pmatrix} \begin{pmatrix} \boldsymbol{i} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{i} \end{pmatrix} \right]^{-1} \begin{pmatrix} \boldsymbol{i}' & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{i}' \end{pmatrix} \\
&= \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{pmatrix} - \begin{pmatrix} \boldsymbol{i} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{i} \end{pmatrix} \begin{pmatrix} T & \boldsymbol{0} \\ \boldsymbol{0} & T \end{pmatrix}^{-1} \begin{pmatrix} \boldsymbol{i}' & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{i}' \end{pmatrix} \\
&= \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{pmatrix} - \begin{pmatrix} \boldsymbol{i} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{i} \end{pmatrix} \begin{pmatrix} 1/T & \boldsymbol{0} \\ \boldsymbol{0} & 1/T \end{pmatrix} \begin{pmatrix} \boldsymbol{i}' & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{i}' \end{pmatrix} \\
&= \begin{pmatrix} \boldsymbol{I} - \frac{1}{T}\boldsymbol{i}\boldsymbol{i}' & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} - \frac{1}{T}\boldsymbol{i}\boldsymbol{i}' \end{pmatrix}
\end{aligned}$$

# $\boldsymbol{M}_D\boldsymbol{y}$ Is the Within-Demeaned Vector

Writing $\bar{y}_i = \frac{1}{T}\sum_t y_{it}$ for unit $i$'s mean:

$$\boldsymbol{M}_D\boldsymbol{y} = \begin{pmatrix} \boldsymbol{I} - \frac{1}{T}\boldsymbol{i}\boldsymbol{i}' & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} - \frac{1}{T}\boldsymbol{i}\boldsymbol{i}' \end{pmatrix} \begin{pmatrix} \boldsymbol{y}_1 \\ \boldsymbol{y}_2 \end{pmatrix} = \begin{pmatrix} y_{11} - \bar{y}_1 \\ \vdots \\ y_{1T} - \bar{y}_1 \\ y_{21} - \bar{y}_2 \\ \vdots \\ y_{2T} - \bar{y}_2 \end{pmatrix}.$$

The $i$th block of $\boldsymbol{M}_D\boldsymbol{y}$ is $\boldsymbol{y}_i$ with unit $i$'s own mean subtracted. The same operation applied column-by-column to $\boldsymbol{X}$ gives $\boldsymbol{M}_D\boldsymbol{X}$. Equivalence to the dummy regression is now confirmed.

# The Within Estimator: Closed Form and Computation

Equivalence confirmed: $\boldsymbol{M}_D \boldsymbol{X}$ and $\boldsymbol{M}_D \boldsymbol{y}$ are the within-demeaned data, so the FE slope is OLS on those demeaned variables:

$$\hat{\beta}_W = (\boldsymbol{S}_{xx}^W)^{-1} \boldsymbol{S}_{xy}^W,$$

$$\boldsymbol{S}_{xx}^W = \sum_{i,t} (\boldsymbol{x}_{it} - \bar{\boldsymbol{x}}_i)(\boldsymbol{x}_{it} - \bar{\boldsymbol{x}}_i)', \qquad \boldsymbol{S}_{xy}^W = \sum_{i,t} (\boldsymbol{x}_{it} - \bar{\boldsymbol{x}}_i)(y_{it} - \bar{y}_i).$$

**In practice.** Demean every variable by its unit mean and run OLS. The residual degrees of freedom shrink from $NT - k$ to $NT - N - k$ to account for the $N$ estimated unit means.

## Stacked Matrix Notation

$\boldsymbol{y}_i$: $T \times 1$, $\boldsymbol{X}_i$: $T \times k$, $\boldsymbol{e}_i$: $T \times 1$, $\beta : k \times 1$

$$\begin{bmatrix} \boldsymbol{y}_1 \\ \boldsymbol{y}_2 \\ \vdots \\ \boldsymbol{y}_N \end{bmatrix} = \begin{bmatrix} \boldsymbol{X}_1 \\ \boldsymbol{X}_2 \\ \vdots \\ \boldsymbol{X}_N \end{bmatrix} \beta + \begin{bmatrix} \alpha_1 \mathbf{1} \\ \alpha_2 \mathbf{1} \\ \vdots \\ \alpha_N \mathbf{1} \end{bmatrix} + \begin{bmatrix} \boldsymbol{e}_1 \\ \boldsymbol{e}_2 \\ \vdots \\ \boldsymbol{e}_N \end{bmatrix}$$

$$y = \boldsymbol{X}\beta + \boldsymbol{\alpha} \otimes \mathbf{1}_T + \boldsymbol{e}$$

# Software Implementations

- lm(y$\sim x + f_1 + f_2 + \ldots + f_n$)
- plm(y$\sim x$, c("unit", "time"))
- fixest::feols(y$\sim$ x$|$ fe)
- You can also demean by time averages, but you need to correct the standard errors.

# Why Demeaning Removes the Unit Effect

The FE model writes

$$y_{it} = \boldsymbol{\beta}' \boldsymbol{x}_{it} + \alpha_i + e_{it}.$$

Subtract each variable's unit mean:

$$y_{it} - \bar{y}_i = \boldsymbol{\beta}'(\boldsymbol{x}_{it} - \bar{\boldsymbol{x}}_i) + (\alpha_i - \bar{\alpha}_i) + (e_{it} - \bar{e}_i).$$

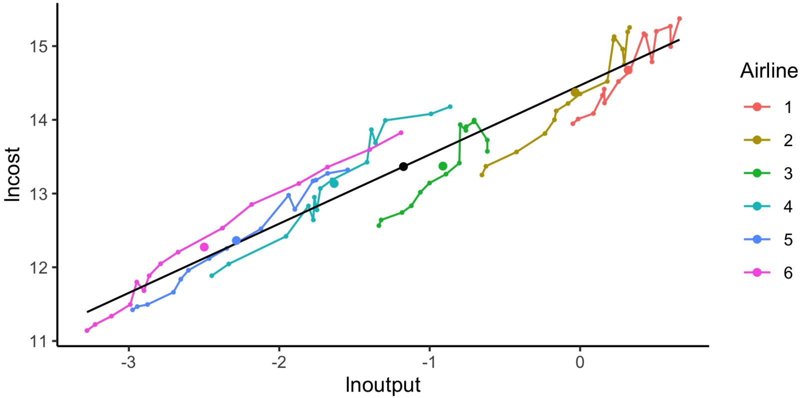
**Key step.** Since $\alpha_i$ does not vary in $t$,

$$\bar{\alpha}_i = \frac{1}{T}\sum_{t=1}^{T} \alpha_i = \alpha_i, \qquad \text{so} \qquad \alpha_i - \bar{\alpha}_i = 0.$$

The unit effect drops out of the demeaned equation. Demeaning removes $\alpha_i$ without our ever observing or estimating it.

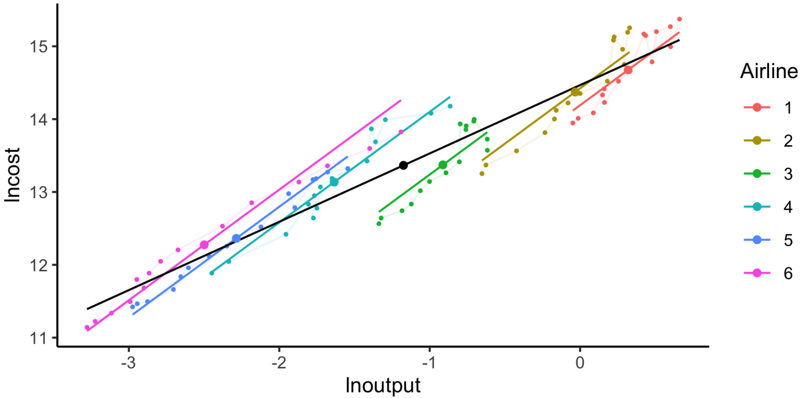
What remains, $\tilde{y}_{it} = \boldsymbol{\beta}' \tilde{\boldsymbol{x}}_{it} + \tilde{e}_{it}$, is what OLS on the demeaned data fits.

# Cost and Output: Pooled Fit




Single black line: pooled OLS slope $\hat{\beta} = 0.88$, averaging across and within airlines.

# Cost and Output: Within Fit




Per-airline fits with a common slope but separate intercepts. Within slope $\hat{\beta} = 0.92$ is steeper than pooled because the flatter between-airline relationship pulls the pooled line down.

# Small note on estimation

$$\boldsymbol{b}_{fe} = \beta + (\boldsymbol{X}'\boldsymbol{M}_D\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{M}_D\boldsymbol{e}$$

Under iid errors, we have: $var(\boldsymbol{b}_{fe}) = \sigma_e^2(\boldsymbol{X}'\boldsymbol{M}_D\boldsymbol{X})^{-1}$. But we need to correct our estimator for the $N$ estimated means:

$$s_e^2 = \frac{\sum_{i=1}^{N}\sum_{t=1}^{T} e_{it}^2}{NT - N - K}$$

# Fixed Effects as GMM Moment Conditions

- Define the within-transformed variables $\tilde{Y}_{it} = Y_{it} - \bar{Y}_i$ and $\tilde{\boldsymbol{X}}_{it} = \boldsymbol{X}_{it} - \bar{\boldsymbol{X}}_i$.
- The FE (within) estimator solves the moment conditions:

$$E[\tilde{\boldsymbol{X}}_{it}' \tilde{e}_{it}] = \mathbf{0}$$

- This is a **just-identified** GMM problem: $k$ moment conditions for $k$ parameters.
  - No overidentifying restrictions $\Rightarrow$ no $J$-test with FE alone.
- **Preview:**
  - Random effects adds between-group moments $\Rightarrow$ more efficient under stronger assumptions.
  - Arellano-Bond (next lecture) creates *many* moment conditions from lagged levels $\Rightarrow$ overidentification $\Rightarrow$ testable with $J$-statistic.

# Pooled Regression Anatomy

Rewrite pooled OLS in the notation we will reuse for between and within.

- Let $\boldsymbol{x}_{it}$ be the $k \times 1$ vector of regressors. Grand means: $\bar{\bar{\boldsymbol{x}}} \equiv \frac{1}{NT} \sum_{i,t} \boldsymbol{x}_{it}$, $\bar{\bar{y}} \equiv \frac{1}{NT} \sum_{i,t} y_{it}$.
- The pooled estimator: $\boldsymbol{b_P} = (\boldsymbol{S}_{xx}^{P})^{-1} \boldsymbol{S}_{xy}^{P}$ with

$$\boldsymbol{S}_{xx}^{P} = \sum_{i,t} (\boldsymbol{x}_{it} - \bar{\bar{\boldsymbol{x}}})(\boldsymbol{x}_{it} - \bar{\bar{\boldsymbol{x}}})', \qquad \boldsymbol{S}_{xy}^{P} = \sum_{i,t} (\boldsymbol{x}_{it} - \bar{\bar{\boldsymbol{x}}})(y_{it} - \bar{\bar{y}}).$$

```
airlines<-airlines%>% mutate(
  mlnoutput = mean(lnoutput),
  mlnpfuel = mean(lnpfuel),
  mload = mean(load))

Xp<-airlines%>%transmute(
                lnoutputdm = lnoutput-mlnoutput,
                lnpfueldm = lnpfuel-mlnpfuel,
                loaddm = load-mload)%>%
                as.matrix()

Sp_xx = t(Xp)%*%Xp
```

## Between Regression

Within threw away all cross-unit variation; **between** does the opposite: it uses only cross-unit variation.

**Construction.** Collapse each unit's $T$ observations into one point at its means; run OLS on the resulting $N$-observation cross section:

$$\bar{y}_i = \beta' \bar{\boldsymbol{x}}_i + \bar{e}_i, \qquad i = 1, \ldots, N.$$

The slope is $\boldsymbol{b}_B = (\boldsymbol{S}^B_{xx})^{-1} \boldsymbol{S}^B_{xy}$ with

$$\boldsymbol{S}^B_{xx} = \sum_{i=1}^{N} T\,(\bar{\boldsymbol{x}}_i - \bar{\bar{\boldsymbol{x}}})(\bar{\boldsymbol{x}}_i - \bar{\bar{\boldsymbol{x}}})', \qquad \boldsymbol{S}^B_{xy} = \sum_{i=1}^{N} T\,(\bar{\boldsymbol{x}}_i - \bar{\bar{\boldsymbol{x}}})(\bar{y}_i - \bar{\bar{y}}).$$

The factor of $T$ keeps these sums on the same scale as $\boldsymbol{S}^P$ and $\boldsymbol{S}^W$, so $\boldsymbol{S}^P = \boldsymbol{S}^B + \boldsymbol{S}^W$ on the next slide.

```
airlines<-airlines%>%
  group_by(airline)%>%
  mutate(
  gmlnoutput=mean(lnoutput),
  gmlnpfuel=mean(lnpfuel),
  gmload=mean(load))

Xb<-airlines%>%transmute(
                blnoutputdm = gmlnoutput-mlnoutput,
                 blnpfueldm = gmlnpfuel-mlnpfuel,
                 bloaddm = gmload-mload) %>%
                 as.matrix()

  Sb_xx = t(Xb)%*%Xb
```

```
airlines<-airlines%>%
  group_by(airline)%>%
  mutate(
  gmlnoutput=mean(lnoutput),
  gmlnpfuel=mean(lnpfuel),
  gmload=mean(load))

  Xw<-airlines%>%transmute(
                plnoutputdm = lnoutput-gmlnoutput,
                 plnpfueldm = lnpfuel-gmlnpfuel,
                 ploaddm = load-gmload)%>%
                 as.matrix()

  Sw_xx = t(Xw)%*%Xw
```

# Decomposition of Variance

$$
\begin{aligned}
y_{it} &= \bar{y}_i + (y_{it} - \bar{y}_i) \\
\boldsymbol{S}_{xx}^{P} &= \boldsymbol{S}_{xx}^{B} + \boldsymbol{S}_{xx}^{W} \\
\boldsymbol{S}_{xy}^{P} &= \boldsymbol{S}_{xy}^{B} + \boldsymbol{S}_{xy}^{W}
\end{aligned}
$$

# Pooled as a Weighted Average of Within and Between

$$
\begin{aligned}
\boldsymbol{b}_P &= (\boldsymbol{S}_{xx}^{P})^{-1}\boldsymbol{S}_{xy}^{P} \\
&= (\boldsymbol{S}_{xx}^{B} + \boldsymbol{S}_{xx}^{W})^{-1}(\boldsymbol{S}_{xy}^{B} + \boldsymbol{S}_{xy}^{W}) \\
&= (\boldsymbol{S}_{xx}^{B} + \boldsymbol{S}_{xx}^{W})^{-1}\boldsymbol{S}_{xy}^{W} + (\boldsymbol{S}_{xx}^{B} + \boldsymbol{S}_{xx}^{W})^{-1}\boldsymbol{S}_{xy}^{B} \\
&= (\boldsymbol{S}_{xx}^{B} + \boldsymbol{S}_{xx}^{W})^{-1}\boldsymbol{S}_{xx}^{W}\boldsymbol{b}_W + (\boldsymbol{S}_{xx}^{B} + \boldsymbol{S}_{xx}^{W})^{-1}\boldsymbol{S}_{xx}^{B}\boldsymbol{b}_B \\
&= (\boldsymbol{S}_{xx}^{B} + \boldsymbol{S}_{xx}^{W})^{-1}\boldsymbol{S}_{xx}^{W}\boldsymbol{b}_W + (\boldsymbol{S}_{xx}^{B} + \boldsymbol{S}_{xx}^{W})^{-1}(\boldsymbol{S}_{xx}^{B} + \boldsymbol{S}_{xx}^{W} - \boldsymbol{S}_{xx}^{W})\boldsymbol{b}_B \\
&= (\boldsymbol{S}_{xx}^{B} + \boldsymbol{S}_{xx}^{W})^{-1}\boldsymbol{S}_{xx}^{W}\boldsymbol{b}_W + (\boldsymbol{I} - (\boldsymbol{S}_{xx}^{B} + \boldsymbol{S}_{xx}^{W})^{-1}\boldsymbol{S}_{xx}^{W})\boldsymbol{b}_B \\
&= \boldsymbol{\lambda}\boldsymbol{b}_W + (\boldsymbol{I} - \boldsymbol{\lambda})\boldsymbol{b}_B
\end{aligned}
$$

Three estimators on the same data, differing only in which variation they use. The next slide shows them side by side on the airline panel.

# Pooled, Between, and Within: Airline Estimates

| | lncost | | |
|---|---|---|---|
| | Pooled | Between | Within |
| lnoutput | 0.883*** | 0.782** | 0.919*** |
| | (0.013) | (0.109) | (0.030) |
| lnpfuel | 0.454*** | −5.524 | 0.417*** |
| | (0.020) | (4.479) | (0.015) |
| load | −1.628*** | −1.751 | −1.070*** |
| | (0.345) | (2.743) | (0.202) |
| Constant | 9.517*** | 85.809 | |
| | (0.229) | (56.483) | |
| Observations | 90 | 6 | 90 |
| $R^2$ | 0.988 | 0.994 | 0.993 |

# What the estimates tell us

Reading the pooled column as Cobb-Douglas elasticities:

- **Output elasticity**

$$\hat{\beta}_1 = 0.883 \text{ (pooled)} \implies \hat{r} = \frac{1}{\hat{\beta}_1} \approx 1.13.$$

  The industry exhibits *mild increasing returns to scale* (costs rise less than proportionally with output).

- **Fuel price elasticity**

$$\hat{\beta}_2 = 0.454 \implies \hat{\alpha}_F = \hat{r}\,\hat{\beta}_2 \approx 1.13 \times 0.454 \approx 0.51.$$

  Roughly half of total variable cost is attributable to fuel.

- **Load factor effect** The coefficient on *load* ($-1.63$) means that a 1-percentage-point increase in capacity utilisation lowers total cost by about 1.6 %, consistent with spreading fixed operating expenses over more output.

# From Cost Functions to Political Economy

- The airline panel illustrated the mechanics: removing unit heterogeneity, pooling within and between variation.
- The same machinery answers questions where the unit effect is the central concern, not a nuisance.
- Next: a country-year panel asking why top marginal tax rates rose so dramatically in the 20th century.

# Political Science Application: Tax Progressivity and War

- **Scheve & Stasavage (2010, 2012):** Why did top tax rates rise dramatically in the 20th century?
- **Hypothesis:** Mass military mobilization during WWI/WWII created political pressure for progressive taxation as a "conscription of wealth."
- **Data:** Country-year panel, 20 countries, 1850–1970.
  - $Y_{it}$: top marginal income tax rate
  - $X_{it}$: `wwihighmobaft` = high-mobilization country $\times$ post-WWI
  - $\alpha_i$: country fixed effects (institutions, fiscal capacity, colonial legacies)
- **Design.** A within-country DiD: did high-mobilization countries see larger jumps in top rates after the war than low-mobilization countries? Country FE absorb time-invariant differences; the identifying assumption is parallel trends across the two groups absent the war.
- **Inference.** Newey-West SEs (next slide) handle within-country serial correlation.

# Scheve & Stasavage: R Implementation

```
library(plm); library(haven); library(fixest)
dta <- read_dta("progressTax.dta")
pdta <- subset(dta, year >= 1900 & year <= 1930)

# Fixed effects with plm
fe_mod <- plm(topratep ~ wwihighmobaft,
              data = pdta,
              index = c("ccode", "year"),
              model = "within")
summary(fe_mod)

# Equivalent with fixest (+ Newey-West SEs)
fe_mod2 <- feols(topratep ~ wwihighmobaft +
    gdppcp + leftseatshp + munsuff + ratiop
    | country,
    data = pdta, panel.id = ~country + year,
    vcov = NW(1) ~ country + year)
```

# Where We Are: From GMM to Panel Data

- In Lectures 15–16 we developed **GMM** as a unified framework:
  - Specify moment conditions $E[\boldsymbol{g}(\boldsymbol{z}_i, \boldsymbol{\theta}_0)] = \boldsymbol{0}$
  - Estimate $\hat{\boldsymbol{\theta}}$ by minimizing $\hat{\boldsymbol{g}}' \boldsymbol{W} \hat{\boldsymbol{g}}$
  - Test overidentifying restrictions with the $J$-statistic
- **Panel data** introduces new structure:
  - Repeated observations create natural moment conditions
  - Unobserved we are dealing with heterogeneity in $\alpha_i$

# Panel Estimators as Moment Conditions

| Estimator | Moment Condition | Key Assumption |
| --- | --- | --- |
| Pooled OLS | $E[\boldsymbol{X}'_{it} e_{it}] = \mathbf{0}$ | $\alpha_i$ uncorrelated with $\boldsymbol{X}$ |
| Fixed Effects | $E[\tilde{\boldsymbol{X}}'_{it} \tilde{e}_{it}] = \mathbf{0}$ | Strict exogeneity (within-unit) |
| Random Effects | $E[\boldsymbol{X}'_{it} \nu_{it}] = \mathbf{0}$ | $\alpha_i$ uncorrelated with $\boldsymbol{X}$; GLS weighting |
| Arellano-Bond | $E[Y_{is} \cdot \Delta e_{it}] = 0, \ s \leq t-2$ | Sequential exogeneity; lagged levels as instruments |

- Each row is a **GMM estimator** with different instruments and assumptions.
- More moment conditions $\Rightarrow$ overidentification $\Rightarrow$ testable with the $J$-test.

# When DiD Fails: Synthetic Control

- DiD recovers $D$ only under parallel trends. For aggregate units (states, countries), one untreated average is rarely a credible counterfactual.
- Idea (Abadie & Gardeazabal 2003; Abadie, Diamond & Hainmueller 2010): match the treated unit to a *weighted average* of control units in the pre-period; the same weights deliver the post-period counterfactual.
- Athey & Imbens (2017) call it "arguably the most important innovation in the evaluation literature in the last fifteen years."

# The Weighting Problem

**Setup.** Treated unit $i = 0$, controls $j = 1, \ldots, J$; pre-period $t \in \mathcal{T}_0$, post-period $t \in \mathcal{T}_1$.

**Find** weights $W = (w_1, \ldots, w_J)$ with $w_j \geq 0$, $\sum_j w_j = 1$ that minimise pre-period fit error:

$$\hat{W} = \arg\min_{W} \sum_{t \in \mathcal{T}_0} \left( Y_{0t} - \sum_j w_j Y_{jt} \right)^2.$$

**Counterfactual and effect:**

$$\hat{Y}_{0t}^0 = \sum_j \hat{w}_j Y_{jt}, \qquad \hat{\alpha}_{0t} = Y_{0t} - \hat{Y}_{0t}^0 \quad (t \in \mathcal{T}_1).$$

**What it generalises.** Linear factor model $Y_{it} = \alpha_i D_{it} + \lambda_i' \mu_t + \epsilon_{it}$: SCM lets each unit have its own multi-dimensional loading $\lambda_i$. DiD assumes a single additive unit effect; SCM handles heterogeneous responses to common factors.

# Two Strong Assumptions

## Convex hull

The treated unit's pre-period outcomes lie inside the convex hull of the controls' pre-period outcomes.

## Perfect pre-period fit

There exist weights $W$ such that the pre-period gap $Y_{0t} - \sum_j w_j Y_{jt}$ is zero for every $t \in \mathcal{T}_0$.

Both routinely fail. Ferman & Pinto (2021): when fit is imperfect, the SCM weights are biased and the bias does *not* vanish as $T_0 \to \infty$ if $J$ is fixed. The mechanism is the same attenuation that plagues regression with measurement-error in regressors.

# Recent Fixes

- **Powell (2018, RAND WR-1246).** Build a synthetic control for *every* unit, not just the treated. If $\hat{w}_{ij}$ puts weight 0.4 on the treated unit when matching unit $i$, then unit $i$'s post-period gap is informative about the treatment effect ($-0.4\times$ it). Also: smooth outcomes before fitting weights to limit attenuation from idiosyncratic noise.
- **Fry (2024, J. Econometrics).** GMM-based SC: split controls into "SC units" and "instruments." Excluded units instrument for included ones; exclusion holds when idiosyncratic shocks are uncorrelated across units. Asymptotically unbiased as $T_0 \to \infty$ with $J$ fixed, even without perfect fit.
- Both relax the perfect-fit assumption that drives Ferman-Pinto bias.

# Application: Massachusetts 2006 Health Reform

- **Question** (Powell 2018): did the 2006 Massachusetts reform (the ACA prototype) reduce mortality?
- **Why not DiD with the national average?** Massachusetts mortality does not track the national average in the pre-period, so the parallel-trend counterfactual is implausible.
- **SCM construction:** weighted average of other states fit to pre-2006 Massachusetts mortality and demographics.
- **Result:** the imperfect-SCM estimator gives a $\approx 3\%$ reduction in the Massachusetts mortality rate. A propensity-score DiD design (Sommers et al. 2014) gives comparable directional results, but the SC fit is dramatically better in the pre-period.

# Summary

- Difference in difference cancels out group-invariant trends and time invariant group differences (whether of interest or not).
- Fixed effects give the same estimates as demeaning $\boldsymbol{X}$ and $\boldsymbol{y}$ by their over time means, the "within" estimator (so long as you correct the residual variance estimator).
- The within estimator corresponds to the GMM moment condition $E[\tilde{\boldsymbol{X}}_{it}' \tilde{e}_{it}] = \mathbf{0}$, and is just-identified.
- A "between" estimator only uses variation across units, averaged over time.
- Pooled Regression is a weighted average of the between and within estimates, with weights $(\boldsymbol{S}_{xx}^{B} + \boldsymbol{S}_{xx}^{W})^{-1}\boldsymbol{S}_{xx}^{W}$.
- **Next time:** Random effects adds between-group moments via GLS, and Arellano-Bond uses lagged levels as instruments in first-differenced dynamic panels. Both fit into the GMM framework from Lectures 15–16.